

# Calculus I

## Continuity

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# Outline

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# Continuity

- Let  $f$  be a function and  $a$  be a point in its domain.
- Suppose  $\lim_{x \rightarrow a} f(x)$  exists.

## Definition (Continuous at a Number)

We say that  $f$  is continuous at  $a$  if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

## Definition (Discontinuous at a Number)

Suppose  $f$  is defined at  $a$ . We say  $f$  is discontinuous at  $a$  if it is not continuous at  $a$ .

Physical phenomena are often continuous. The majority of the physical phenomena that are understood are continuous. Examples:

- Motion of a vehicle with respect to time without sudden brakes.
- Orbits of planets and celestial bodies with respect to time.
- A person's height with respect to time.
- And many more.

Discontinuous phenomena examples:

- Particle velocities during collisions and explosions.
- Electric current phenomena, gating events in porins (the event of a molecule passing in and out of a cell).
- Particle physics phenomena.
- And many more.

## Example

The picture below shows a graph of a function  $f$ . At which numbers is  $f$  either discontinuous or not defined? Why?

- Not defined at 1:
- $\lim_{x \rightarrow 1} f(x)$  exists.
- $f(1)$  is not defined.
- Discontinuous at 2:
- $f(2)$  is defined.
- $\lim_{x \rightarrow 2} f(x)$  doesn't exist.
- Discontinuous at 4:
- $f(4)$  is defined.
- $\lim_{x \rightarrow 4} f(x)$  exists.
- $\lim_{x \rightarrow 4} f(x) \neq f(4)$ .

## Definition (Greatest Integer Function)

The *greatest integer function*  $\lfloor x \rfloor$  is defined as the largest integer that is less than or equal to  $x$ .

In computer science this function is called the *floor* function.

$$\lfloor 4 \rfloor = 4$$

$$\lfloor 4.8 \rfloor = 4$$

$$\lfloor \pi \rfloor = 3$$

$$\lfloor \sqrt{2} \rfloor = 1$$

$$\left\lfloor -\frac{1}{2} \right\rfloor = -1$$

$$\lfloor -\pi \rfloor = -4$$

## Example

Where is this function discontinuous?

$$f(x) = \begin{cases} \frac{x^2-x-2}{x-2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

- $f(2)$  is defined ( $f(2) = 1$ ).
- $\lim_{x \rightarrow 2} f(x)$  exists ( $= 3$ ).
- $\lim_{x \rightarrow 2} f(x) \neq f(2)$ .
- Discontinuous at 2.
- This is called a removable discontinuity because we can redefine  $f$  at one point to make  $f$  continuous.

## Example

Where is this function discontinuous?

$$f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

- $f(0)$  is defined ( $f(0) = 1$ ).
- $\lim_{x \rightarrow 0} f(x)$  doesn't exist ( $\infty$ ).
- Discontinuous at 0.
- This is called an infinite discontinuity.

## Example

Where is this function discontinuous?

$$f(x) = \lfloor x \rfloor$$

- $f(1)$  exists ( $f(1) = 1$ ).
- $\lim_{x \rightarrow 1^+} f(x) = 1$ .
- $\lim_{x \rightarrow 1^-} f(x) = 0$ .
- $\lim_{x \rightarrow 1} f(x)$  doesn't exist.
- Discontinuous at 1.
- Discontinuous at every integer  $n$ .
- The left and right limits both exist but are not equal.
- Such discontinuities are called jump discontinuities (the function appears to “jump”).

## Definition (Continuous from the Right or Left)

A function  $f$  is continuous from the right at a number  $a$  if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

and  $f$  is continuous from the left at  $a$  if

$$\lim_{x \rightarrow a^-} f(x) = f(a).$$

## Example

Consider  $f(x) = \lfloor x \rfloor$ , and pick any integer  $n$ .

- $f(n) = n$ .
- $\lim_{x \rightarrow n^+} f(x) = n$ .
- Continuous from the right at  $n$ .
- $\lim_{x \rightarrow n^-} f(x) = n - 1$ .
- Discontinuous from the left at  $n$ .

## Definition (Continuous on an Interval)

A function  $f$  is continuous on an interval if it is continuous at every number in the interval.

- If  $f$  is defined at the **right left** endpoint of an interval, continuous means continuous from the **left right**.
- Think of a function that is continuous on an interval as a function that has no breaks in its graph, and so can be drawn “without lifting your pen”.

## Theorem (Algebra of Continuous Functions)

*If  $f$  and  $g$  are continuous at  $a$  and  $c$  is a constant, then the following functions are also continuous at  $a$ :*

①  $f + g$

③  $cf$

⑤  $\frac{f}{g}$  if  $g(a) \neq 0$ .

②  $f - g$

④  $fg$

### Proof.

$$\lim_{x \rightarrow a} f(x) = f(a) \text{ and } \lim_{x \rightarrow a} g(x) = g(a).$$

$$\begin{aligned} \lim_{x \rightarrow a} (f + g)(x) &= \lim_{x \rightarrow a} [f(x) + g(x)] \\ &= \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) \quad (\text{by Law 1}) \\ &= f(a) + g(a) = (f + g)(a) \end{aligned}$$

This shows  $f + g$  is continuous at  $a$ . The other parts are similar.  $\square$

## Theorem (Classes of Continuous Functions)

*The following types of functions are continuous at every number in their domains:*

*polynomials      rational functions*  
*root functions      trigonometric functions*

## Theorem (Compositions of Continuous Functions)

*If  $g$  is continuous at  $a$  and  $f$  is continuous at  $g(a)$ , then the composition function  $f \circ g$  given by  $(f \circ g)(x) = f(g(x))$  is continuous at  $a$ .*

## Example

Find  $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$ .

The function  $f(x) = \frac{x^3 + 2x^2 - 1}{5 - 3x}$  is rational, so is continuous on its domain. Its domain is given by  $x \neq \frac{5}{3}$ .

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} &= \lim_{x \rightarrow -2} f(x) \\ &= f(-2) \\ &= \frac{(-2)^3 + 2(-2)^2 - 1}{5 - 3(-2)} \\ &= -\frac{1}{11} \end{aligned}$$

## Example

Where is the function  $F(x) = \frac{1}{\sqrt{x^2+7}-4}$  continuous?

- We can write  $F$  as the composition of 4 functions:
- $F = f \circ g \circ h \circ k$ , or  $F(x) = f(g(h(k(x))))$ .
- $k(x) = x^2 + 7$ .
- $h(u) = \sqrt{u}$ .
- $g(v) = v - 4$ .
- $f(w) = \frac{1}{w}$ .
- These functions are continuous on their domains, so  $F$  is continuous on its domain.
- Its domain is given by  $x \neq 3$  and  $x \neq -3$ .
- Therefore  $F$  is continuous on  $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$ .

## Theorem (The Intermediate Value Theorem)

*Suppose  $f$  is continuous on the closed interval  $[a, b]$  and let  $N$  be any number between  $f(a)$  and  $f(b)$ , where  $f(a) \neq f(b)$ . Then there exists a number  $c$  in  $(a, b)$  such that  $f(c) = N$ .*

## Example

Show that there is a root of the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$

between 1 and 2.

- Let  $f(x) = 4x^3 - 6x^2 + 3x - 2$ .
- $f$  is continuous.
- Use the IVT with  $a = 1$ ,  $b = 2$ , and  $N = 0$ .
- $f(1) = -1$ .
- $f(2) = 12$ .
- $f(1) < 0 < f(2)$ .
- Therefore there is a  $c$  between 1 and 2 such that  $f(c) = 0$ .